

Crusher to the Bridge: A Mechanistic Monograph on Shipboard
Outbreak Simulation, Intervention-Shadowed Calibration, and
Surveillance Saturation

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Abstract

Cruise-ship outbreak observations are not neutral samples from uncontrolled epidemics. They are produced by a coupled process in which passengers become ill, syndromic cases report to medical staff, regulatory thresholds activate intensified sanitation and isolation, and only then do public datasets record the event. This monograph rewrites the CTB evidence around that measurement problem. The main result is that the CDC Vessel Sanitation Program (VSP) norovirus attack-rate distribution is best treated as an intervention-shadowed observation. A single calibrated norovirus shedding parameter, `dose_adj`=10.6, produces high and platform-dependent attack rates under no response: 23.4% on the expedition platform, 22.6% on the classic platform, 12.2% on the spirit platform, and 10.6% on the mega platform. With syndromic detection and VSP-like response, the same parameter yields the observed 5–7% band: 6.7%, 7.1%, 4.6%, and 5.5%, respectively.

The same intervention-shadowed logic explains the surveillance experiments. Once an onboard response is active, marginal gains from wearables, wastewater triggers, and multiplex testing are small relative to the first syndromic trigger. In no-response runs, mean attack rates are 17.2% for norovirus, 20.7% for SARS-CoV-2, 37.9% for measles in a susceptible population, 66.1% for influenza, and 12.0% for *C. difficile*. Under basic syndromic response, these fall to 5.9%, 0.4%, 1.3%, 1.9%, and 0.2%. Advanced mid-voyage surveillance usually changes absolute attack rate by less than one percentage point for norovirus and by similarly small values for respiratory pathogens already contained by isolation. Paired native-transport and CONTAM validation runs show zero infection-outcome deltas despite environmental mass-scale differences, supporting native transport for attack-rate experiments while requiring separate calibration for environmental surveillance thresholds. A stylized starship demonstration shows that the framework generalizes to arbitrary confined environments, but it is secondary to the cruise-ship calibration and surveillance claims.

Keywords: agent-based model; maritime epidemiology; norovirus; cruise ship; Vessel Sanitation Program; outbreak response; CONTAM; surveillance; confined populations

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Chapter 1

Introduction

1.1 Claim: cruise-ship outbreak data are measurements of a response system

Cruise ships are useful confined-population laboratories because they combine repeated shared meals, dense social spaces, cabin-level residence, crew work networks, shared environmental infrastructure, and an onboard medical service operating under external reporting rules. They are also open-boundary systems. Boarding, port calls, shore excursions, provisioning, crew turnover, and disboarding import and export risk. A shipboard model therefore cannot be judged only by whether it reproduces a single observed attack-rate distribution. It must reproduce the process that produces the observation.

The central claim of this monograph is that the observed cruise-ship norovirus attack-rate distribution, especially the CDC Vessel Sanitation Program (VSP) distribution, is an intervention-shadowed measurement. A reportable acute gastroenteritis event is observed after symptomatic passengers have appeared, sick-call behavior has exposed the event to medical staff, sanitation and isolation protocols have changed the onboard environment, and a threshold reporting rule has selected the events that enter the public record [1–5]. Treating those records as uncontrolled epidemic targets biases calibration. The more defensible target is the joint output of biology, ship structure, passenger behavior, and VSP-like response.

Crusher to the Bridge (CTB) was built to evaluate that joint process. The model represents individual agents moving through spatially explicit vessels, receiving pathogen dose through route-weighted transmission pathways, and entering medical response workflows. It was applied to four cruise-platform classes, ten real-world pathogens, paired transport-solver validation, multiple surveillance strategies, and a smaller arbitrary-platform demonstration. The argument is not that the model is a complete digital twin of a cruise line. It is that a mechanistic model with explicit observation and response logic resolves an apparent contradiction in the empirical record: uncontrolled physical platform differences can be large, while observed VSP attack rates remain

compressed.

1.2 Cruise vessels as epidemiological systems

The cruise-ship setting has three features that make it scientifically informative. First, shipboard populations are spatially bounded for days at a time. Dining rooms, buffets, theaters, corridors, cabins, crew messes, medical centers, and air-handling zones constrain where contacts occur. Second, the same people repeat routines. Meal timing, cabin occupancy, crew shifts, and entertainment schedules produce recurrent exposures rather than a fully mixed population. Third, onboard medical teams can alter transmission while the outbreak is still unfolding. Cabin isolation, heightened cleaning, conversion from self-service to staff-served food, diagnostic testing, and port-health reporting are part of the disease system, not external annotations [1, 5, 6].

The literature shows that these systems can support gastrointestinal, respiratory, and rarer high-consequence outbreaks. Norovirus has long dominated reported cruise-associated acute gastroenteritis [2, 3, 7–9]. SARS-CoV-2 outbreaks, including Diamond Princess, showed how fast respiratory pathogens can exploit shared cabins, social mixing, delayed case recognition, and operational constraints [10–16]. Other shipboard events involving influenza, measles, Legionella, and hantavirus demonstrate that the vessel is a platform for multiple route classes rather than a norovirus-only environment [17–22].

1.3 Why calibration against observed attack rates is not enough

An attack rate is a cumulative outcome, but it does not reveal whether transmission stopped because biology was weak, contacts were sparse, susceptibility was low, medical detection was fast, or a regulatory backstop activated. This distinction matters most for norovirus. The parsed VSP dataset used here contains 285 outbreak records from 1993–2023, including 211 coded as norovirus. Norovirus records cluster around a 5–7% passenger attack-rate band for most passenger-count bins, but the VSP threshold itself is near 3%. The public record therefore starts near the point where operational escalation begins.

This monograph treats calibration as a coupled inference problem. A model can match observed VSP attack rates for the wrong reason if it lowers uncontrolled transmissibility until every platform sits near 5–7%. That approach erases real physical differences in dining topology, venue density, and response timing. The revised calibration asks a different question: can one biological shedding parameter create high, platform-dependent uncontrolled epidemics while the modeled VSP response compresses the observed distribution? The answer is yes for the available CTB data.

1.4 Structure of the monograph

Chapter 2 formalizes the CTB architecture, including dose response, route-weighted dose aggregation, density scaling, VSP threshold logic, and cascade surveillance. Chapter 3 defines vessel classes and operational parameterization. Chapter 4 explains the pathogen profiles and why route weights change platform sensitivity. Chapter 5 develops the VSP-shadow argument and presents the norovirus calibration result. Chapter 6 validates the native transport solver against CONTAM for infection outcomes. Chapter 7 evaluates onboard detection and intervention strategies. Chapter 8 gives a short arbitrary-platform demonstration using Starfleet scenarios. Chapters 9 and 10 synthesize implications and future work. The supplement contains real tables for platform assumptions, pathogen profiles, surveillance definitions, campaign provenance, VSP summaries, calibration estimates, intervention effects, transport validation, and source datasets.

Chapter 2

Model Architecture

2.1 Claim: CTB is an observation-and-response model, not only a transmission model

CTB is an epoch-based individual simulation of outbreak dynamics in confined vessels. Each run initializes a platform, a population, a pathogen profile, an immunity assumption, a surveillance strategy, and a random seed. The simulation advances in one-hour epochs over a 168-hour voyage. During each epoch, agents occupy zones, interact through direct and environmental pathways, accumulate pathogen dose, change infection state, and may enter detection or intervention workflows.

The model inherits the Korkin-style individual-state framing but extends it for maritime use with ship-specific zone graphs, per-pathogen route weights, configurable density scaling, dining-venue parameterization, medical response, VSP-like threshold triggers, and a native-versus-CONTAM transport branch [23–27]. The architecture is shown in Figure 2.1. The purpose of these extensions is to model both infection generation and infection observation.

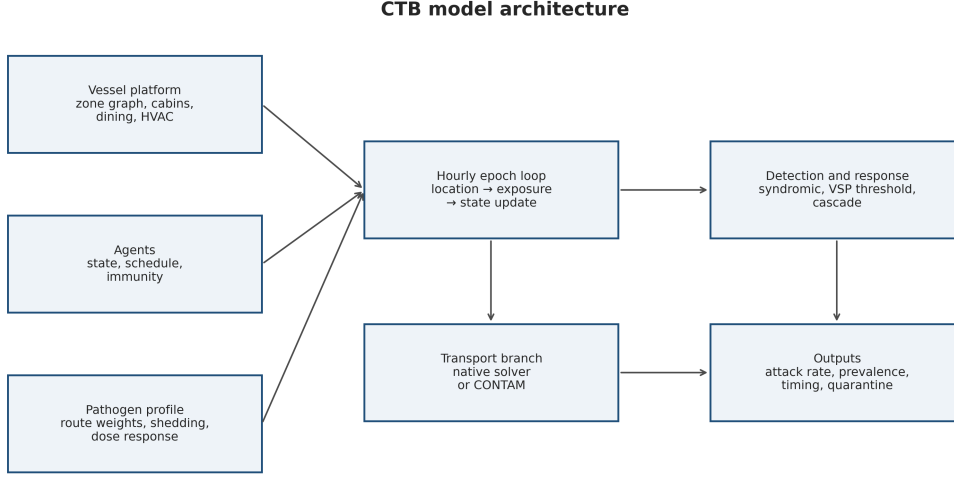


Figure 2.1: CTB architecture. Vessel structure, agent schedules, pathogen profiles, hourly exposure updates, transport solvers, and detection workflows are represented as coupled model components.

2.2 Agent state and event timing

Each agent has a disease state, location schedule, susceptibility modifier, and intervention state. A minimal state set is

$$X_i(t) \in \{S, E, I_s, I_a, R, Q\}, \quad (2.1)$$

where S is susceptible, E exposed, I_s symptomatic infectious, I_a asymptomatic infectious, R recovered or no longer infectious, and Q isolated or quarantined. Not every pathogen uses the same clinical interpretation of these states. The state machine is intended as a common computational scaffold for route-specific profiles.

Transmission is evaluated by epoch. Let $z_i(t)$ be the zone occupied by agent i at epoch t , and let $\mathcal{I}(t)$ be infectious agents. The model computes an exposure dose to susceptible agent i by summing route-specific contributions over infectious agents, environmental reservoirs, and transport fields. The resulting dose is then mapped to infection probability.

2.3 Route-weighted dose aggregation

For pathogen p , CTB represents transmission routes as a vector of non-negative weights $w_{p,r}$ over routes $r \in \mathcal{R}$, including close contact, short-range respiratory, aerosol or HVAC-mediated transport, fomite exposure, food-service exposure, and environmental reservoirs. The route-weighted dose for agent i in epoch t is

$$D_{i,p}(t) = \theta_p \sum_{r \in \mathcal{R}} w_{p,r} d_{i,p,r}(t), \quad (2.2)$$

where $d_{i,p,r}(t)$ is the route-specific dose contribution and θ_p is the calibrated shedding or dose-adjustment factor represented in campaign tables as `dose_adj`. In the available implementation, lower `dose_adj` values generally correspond to higher effective exposure for dose-responsive pathogens, but the mapping is pathogen-specific because each profile has different route weights and susceptibility assumptions.

When several infectious agents occupy a shared setting, dose is density scaled rather than treated as a fixed pairwise constant. The generic contact term is

$$C_z(t) = \beta_z \left(\frac{N_z(t)}{A_z} \right)^\alpha I_z(t), \quad (2.3)$$

where $N_z(t)$ is occupancy, A_z is a zone-size proxy, $I_z(t)$ is infectious occupancy, β_z is a venue-specific contact parameter, and α is the density exponent. The campaign sequence tested α values from 0.50 to 1.00 and used $\alpha = 0.75$ for the main calibrated runs. This exponent permits sublinear crowding effects, which better matched cross-platform behavior than a purely frequency-dependent model in prior CTB calibration work.

2.4 Dose-response mapping

The infection probability for susceptible agent i after epoch dose $D_{i,p}(t)$ follows an exponential dose-response form,

$$P\{S_i(t) \rightarrow E_i(t+1)\} = 1 - \exp[-\kappa_p D_{i,p}(t)], \quad (2.4)$$

with pathogen-specific scale κ_p . For pathogens where a beta-Poisson form is more biologically interpretable, the equivalent profile can be written as

$$P_{\text{inf}}(D) = 1 - \left(1 + \frac{D}{N_{50,p}} (2^{1/\eta_p} - 1) \right)^{-\eta_p}, \quad (2.5)$$

where $N_{50,p}$ is the median infectious dose and η_p controls curvature. The CTB campaign outputs expose the calibrated profile through `dose_adj` and observed attack rates rather than by exporting every underlying dose-response coefficient. The equations define the model logic used to interpret those campaign outputs and align with dose-response framing in norovirus and outbreak-model literature [25, 28, 29].

2.5 VSP threshold logic

VSP-like response is modeled as both an observation rule and an intervention trigger. Let

$$A_{\text{GI}}(t) = \frac{C_{\text{sym,GI}}(t)}{N_{\text{pax}}}, \quad (2.6)$$

where $C_{\text{sym,GI}}(t)$ is the cumulative number of passenger gastrointestinal cases known to the medical system. A VSP threshold τ activates response when

$$A_{\text{GI}}(t) \geq \tau. \quad (2.7)$$

The main VSP-like setting uses $\tau = 0.03$. Sensitivity runs also evaluate 1% and 5% thresholds and a sick-call-only condition in which medical response occurs without the threshold backstop.

Activation changes downstream hazards. It increases case isolation, intensifies sanitation, changes food-service operations toward staff service, and can quarantine contacts. In simplified notation,

$$\beta_{z,r}(t+1) = \beta_{z,r}(t) \times m_r^{\text{response}}, \quad 0 \leq m_r^{\text{response}} \leq 1, \quad (2.8)$$

for routes and zones affected by response. The exact multipliers are implemented in model configuration rather than inferred from the aggregate CSVs.

2.6 Surveillance and cascade logic

CTB distinguishes a direct syndromic trigger from cascade strategies. A syndromic trigger occurs when symptomatic agents report to medical staff with probability p_{sick} after a delay δ_{detect} . Cascade strategies add possible signals from multiplex point-of-care tests, crew or passenger wearable programs, wastewater, or environmental surveillance. For signal type s ,

$$T_s = \min\{t : Y_s(t) \geq h_s\}, \quad (2.9)$$

where $Y_s(t)$ is the observed signal and h_s is the strategy-specific threshold. The intervention time is

$$T_{\text{response}} = \min_s T_s + L, \quad (2.10)$$

where L is decision latency. The surveillance experiments in Chapter 7 show that lowering T_{response} has limited marginal value once the syndromic and VSP mechanisms already trigger early enough to cut transmission.

Chapter 3

Vessel Classes and Operational Parameterization

3.1 Claim: platform differences are structural before they are epidemiological

The four cruise platforms are model classes, not named ships. They represent increasing population size and venue complexity: expedition with 450 agents, classic with 1,910 agents, spirit with 3,000 agents, and mega with 7,000 agents. Their epidemiological differences arise from structural operations: dining architecture, zone modularity, self-service exposure, crew-to-passenger ratios, and medical response gradients. These inputs determine how a biological pathogen profile is translated into contact opportunities and observed cases.

3.2 Dining networks and food-service exposure

Food service is a major driver of shipboard gastrointestinal transmission because it combines repeated attendance, high-throughput queues, shared utensils, crew-mediated service, and common environmental surfaces. The dining source report separates vessels into expedition, premium, contemporary, and mega or iconic tiers. Expedition ships have small passenger counts, high service ratios, and minimal unmonitored self-service. Premium and contemporary vessels combine main dining rooms, lido buffets, specialty dining, and mixed fixed/open seating. Mega ships distribute passengers across many venues, but the absolute number of meals and surface touches is large.

Figure 3.1 summarizes the meal-allocation assumptions used for model interpretation. The strongest contrast is between mediated service and self-service. Expedition-class assumptions place most dinner exposure in the main dining room and make buffet exposure partly crew-served. Contemporary and mega assumptions allocate more daily interactions to casual food halls, distributed

buffets, specialty venues, and open seating. These structures change both direct contact degree and fomite exposure. Literature on cruise-associated acute gastroenteritis and norovirus supports treating dining and shared surfaces as major transmission contexts [1–3, 7–9].

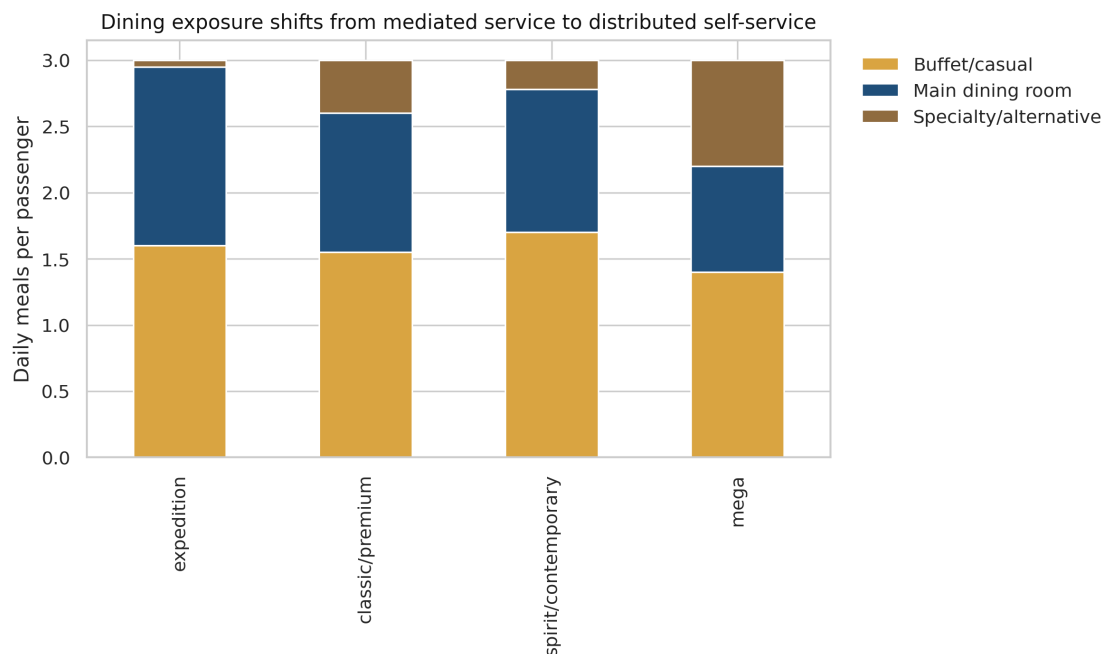


Figure 3.1: Daily meal-allocation assumptions by platform class. Values are meals per passenger per day by venue type, adapted from the project dining parameterization source.

3.3 Medical response gradients

The medical-response source report describes shipboard clinics as small autonomous medical systems. Large vessels typically carry two physicians, four to six nurses, paramedics, and healthcare assistants, while expedition vessels may carry one physician with limited support. Smaller vessels have fewer staff but less anonymity and tighter social knowledge. Larger vessels have more equipment and personnel but a weaker per-capita signal. CTB represents these differences through sick-call probability, detection delay, isolation compliance, and response escalation parameters.

The calibration result in Chapter 5 shows that these medical knobs do not dominate final norovirus attack rate once the VSP threshold backstop is active. That does not make medical operations irrelevant. It means that, for the calibrated norovirus runs, the thresholded response is high leverage enough to swamp moderate variation in sick-call probability, detection delay, and isolation compliance. Medical structure matters most before the backstop activates, at voyage boundaries, and for pathogens whose detection signals differ from gastrointestinal illness.

3.4 Boundary conditions: boarding, ports, and turnaround

The ship-operations specification adds an itinerary layer with boarding, sea days, port days, and disboarding. In the current implemented campaigns, port effects are stubbed by default so the main experiments are effectively all-sea-day voyages. The specification is still important for interpretation. Boarding can force 70–85% of passengers into lido buffets before staterooms open. Port days can remove 60–80% of passengers from the vessel during daytime hours. Turnaround can carry environmental reservoirs or infected crew between voyages.

These processes are boundary conditions rather than mid-voyage onboard response. The Chapter 7 surveillance result points directly at them: if syndromic response already captures most onboard benefit after cases are established, advanced surveillance may produce more value by preventing infectious boarding, detecting terminal wastewater signals, identifying risky turnaround contamination, or supporting early cancellation and offboarding decisions.

3.5 Platform assumptions used in the model

Table [A.1](#) in the supplement lists the platform assumptions used in this revision. The table is intentionally operational. It records population size, dining exposure, medical response default, and boundary-condition status because those are the model features that connect ship design to disease outcomes. Full zone inventories should be generated from the code repository in a later synchronized build, but the available campaign CSVs and context documents support the class-level interpretation used here.

Chapter 4

Pathogen Parameterization

4.1 Claim: route weights are the mechanism that makes one ship model multi-pathogen

A confined vessel is not a single transmission environment. The same corridor, cabin, buffet, or clinic has different meaning for a respiratory virus, a gastrointestinal virus, a spore-forming bacterium, a foodborne bacterium, or a rare high-consequence pathogen. CTB therefore represents each pathogen through a profile rather than through a universal transmissibility scalar. The profile combines route weights, a dose-adjustment parameter, susceptibility or immunity assumptions, incubation and infectiousness timing, and intervention sensitivity.

The real-world pathogen set includes norovirus, SARS-CoV-2, influenza, measles, *Clostridioides difficile*, campylobacter, vibrio, Legionella, Ebola, and hantavirus. The main intervention campaign focuses on five calibrated or policy-relevant examples: norovirus, SARS-CoV-2, influenza, measles, and *C. difficile*. Earlier route-weight sweeps include the broader ten-pathogen set to confirm that the model can separate route classes rather than only rescale one disease curve.

4.2 Route classes

CTB uses six conceptual transmission routes. Close-contact transmission captures direct interpersonal exposure in cabins, dining rooms, crew spaces, clinics, queues, and entertainment venues. Short-range respiratory transmission captures near-field droplets and aerosols within shared zones. HVAC or long-range aerosol transport captures airborne movement through connected zones and is evaluated through the native solver or CONTAM branch. Fomite transmission captures contaminated high-touch surfaces. Food-service transmission captures route-specific exposure through food preparation, buffet utensils, service mode, and dining attendance. Environmental-reservoir transmission captures persistent contamination relevant to pathogens such as norovirus and *C. difficile*.

Norovirus is weighted toward fomite, food-service, direct contact, and environmental routes, consistent with low infectious dose, environmental stability, asymptomatic shedding, and the repeated role of food and shared surfaces in cruise outbreaks [2, 3, 9, 28–30]. SARS-CoV-2, influenza, and measles place more weight on respiratory and aerosol routes, with pathogen-specific differences in overdispersion, susceptibility, and intervention timing [17, 18, 25, 27, 31–33]. Hantavirus and Ebola are treated as high-consequence profiles whose route assumptions are narrower and more uncertain [21, 22, 25, 34]. Legionella is primarily environmental and aerosolized water-system exposure rather than person-to-person spread [19, 20, 35]. Campylobacter and vibrio are food or water associated in the shipboard framing [36–38].

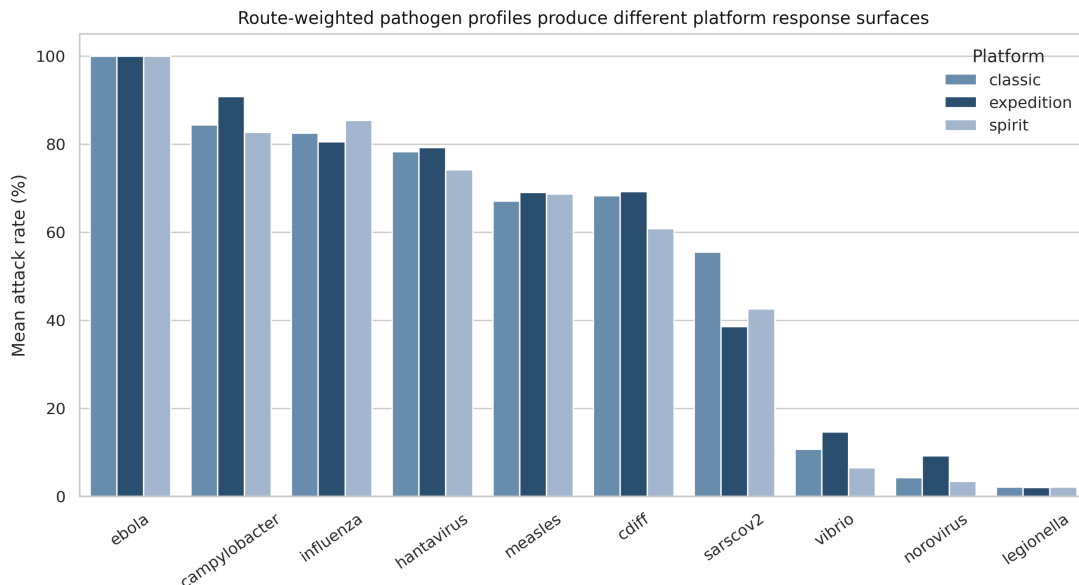


Figure 4.1: Baseline route-weighted pathogen behavior in the broad ten-pathogen sweep. Mean attack rates vary by pathogen and platform because route weights interact with dining topology, density, and venue structure.

4.3 Dose adjustment is a shedding and exposure modulator

Campaign outputs report `dose_adj` as the exposed calibration parameter. It should be interpreted as an effective shedding, exposure, or infectious-dose modulator within a pathogen profile rather than as a directly measured biological constant. In the calibrated norovirus result, `dose_adj`=10.6 is the value that reconciles high uncontrolled platform-specific epidemics with the compressed VSP-observed attack-rate band once the response process is included.

The same numerical `dose_adj` value should not be compared across pathogens as if it were a universal transmissibility index. A respiratory profile with high aerosol weight and short serial interval can respond differently from a gastrointestinal profile with strong environmental persistence. The route-weight formulation is what makes multi-pathogen comparison interpretable.

4.4 Pathogen profiles used for interpretation

The supplement provides the working pathogen-profile table available from campaign outputs and context notes. It reports route emphasis, calibrated campaign dose values, immunity assumptions, and chapter use. The table is not a complete export of every code-level parameter because the supplied artifact set does not include the full pathogen configuration files. That gap is noted explicitly rather than filled with reconstructed constants.

Chapter 5

Calibration and the VSP Shadow

5.1 Claim: the VSP distribution is an intervention-shadowed target

The norovirus calibration problem is the central result of this monograph. VSP acute-gastroenteritis data are often tempting calibration targets because they are numerous, standardized, and ship-associated. They are not, however, uncontrolled epidemic measurements. They are produced after symptom onset, passenger reporting, medical recognition, sanitation escalation, possible isolation, and regulatory reporting [1–6]. The 3% VSP reporting threshold is therefore both an observation rule and a response trigger.

This distinction resolves the apparent platform paradox. If the VSP distribution is treated as an uncontrolled target, model calibration must suppress physical ship-size differences until all platforms sit near 5–7% attack rate. If the VSP distribution is treated as a post-response target, a single norovirus biological parameter can produce larger uncontrolled differences and then compress them through the modeled response system.

5.2 Observed VSP attack rates are compressed

The parsed VSP table contains 285 outbreak rows from 1993–2023 and 211 norovirus-coded rows. Binning norovirus records by passenger count gives mean passenger attack rates of 17.9% for ships under 500 passengers, 6.2% for 500–1500 passengers, 5.9% for 1500–3000 passengers, 6.5% for 3000–5000 passengers, and 8.9% for the single record above 5000 passengers. The small-vessel bin is sparse and unstable; the middle bins form the observed 5–7% band.

Figure 5.1 shows why that band should not be used as a no-response target. At `dose_adj`=10.6, no-response norovirus runs retain a physical platform gap: 23.4% on expedition, 22.6% on classic, 12.2% on spirit, and 10.6% on mega. Under syndromic and VSP-like response variants, the same

dose compresses to 6.7%, 7.1%, 4.6%, and 5.5%.

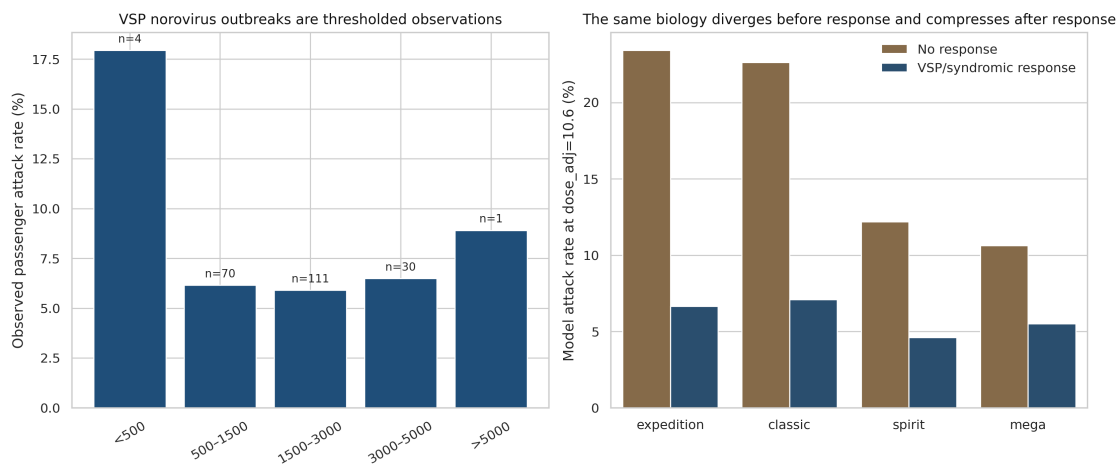


Figure 5.1: The VSP shadow. Left: observed VSP norovirus attack rates are compressed for most passenger-count bins. Right: the same norovirus parameter produces high, platform-dependent no-response attack rates but a compressed VSP-like observed band after response.

5.3 Calibration estimate

Table A.6 reports the calibrated norovirus estimates with 95% confidence intervals over seeds and response variants. The no-response attack-rate ratio between expedition and mega is 2.20 at `dose_adj`=10.6. After VSP-like response, the expedition-to-mega ratio falls to 1.20, and the classic-to-mega ratio is 1.29. The response therefore reduces both mean attack rate and cross-platform dispersion.

This result changes the calibration philosophy. The biological shedding parameter is chosen to match the intervention-shadowed observation, not to erase the uncontrolled gap. The uncontrolled gap remains an estimate of latent epidemic potential under the same biology. That potential is not directly visible in VSP records because VSP records are generated by a system that intervenes.

5.4 The VSP threshold is a high-leverage policy parameter

A threshold sweep confirms that the response trigger is high leverage. For norovirus with no baseline immunity, no response yields platform means ranging from roughly 10.6% to 23.4%. A sick-call-only response reduces the range but leaves more platform spread. VSP-style thresholds progressively compress the distribution: a 5% threshold produces roughly 5.8–7.9%, the 3% threshold produces roughly 5.1–7.0%, and a 1% threshold produces roughly 4.3–6.0% (Figure 5.2).

Medical sensitivity runs vary sick-call probability, detection delay, and isolation compliance. Final attack rates change little once the VSP threshold backstop is active. This does not mean

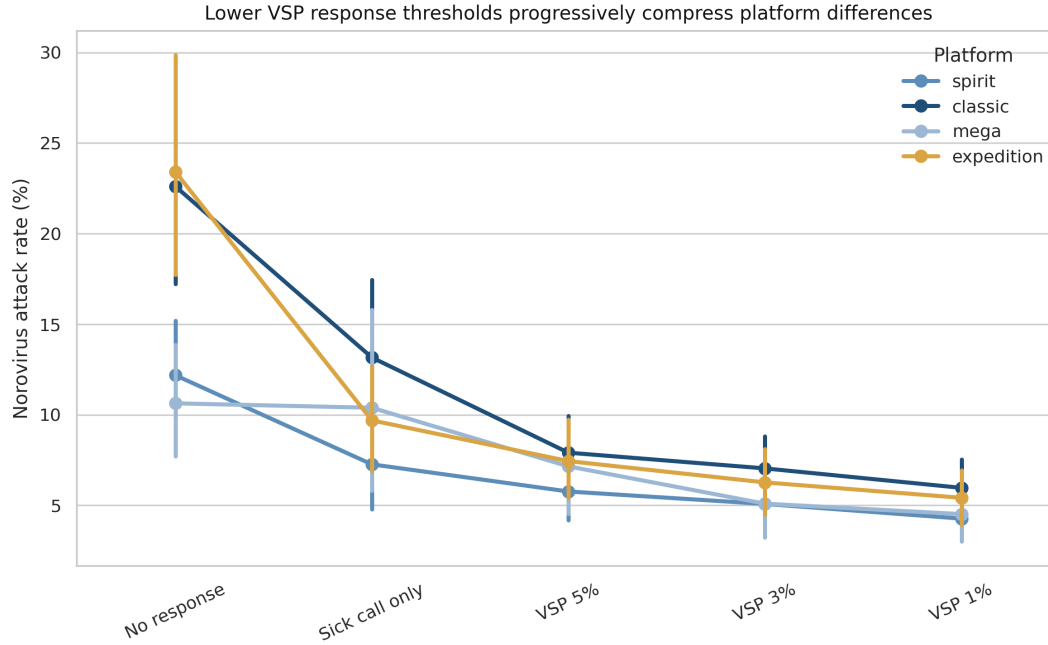


Figure 5.2: Norovirus attack rates under progressively stricter VSP-like response thresholds. Lower thresholds compress platform differences and reduce mean attack rate.

that sick-call behavior or compliance cannot matter in other parameter regimes. It means that, in the calibrated norovirus regime, the thresholded escalation dominates final attack rate.

5.5 Interpretation

The model suggests that the apparent VSP flatness across ship sizes is not evidence that ship structure is irrelevant. It is evidence that the public observation process is already post-intervention. Uncontrolled physical structure still matters, but the VSP-like response masks much of it. This finding is consistent with outbreak-model principles: observed epidemic size is shaped by detection, control, and overdispersion as much as by intrinsic transmissibility [25–27, 39, 40].

Chapter 6

Transport Solver Validation

6.1 Claim: infection outcomes validate across solvers, but environmental mass thresholds do not

CTB includes a native analytical transport solver and a CONTAM branch for multi-zone airflow and contaminant movement. The validation question is not whether the two solvers generate identical environmental mass at every point. It is whether substituting the transport branch changes infection outcomes at the calibrated attack-rate scale. For the available norovirus validation, it does not.

6.2 Paired validation design

The validation compares paired native and CONTAM runs by seed, platform, pathogen profile, and intervention setting. The project notes report 200 paired runs with zero deltas across attack rate, peak prevalence, outbreak occurrence, peak epoch, detection epoch, detection lag, effective reproduction summary, quarantine-person-epochs, and final susceptible fraction. A stricter per-seed, per-epoch norovirus validation reports 6,720 of 6,720 agent-state observations exactly matched. The quick-pair telemetry file rechecked 40 paired runs and likewise recorded zero agent-state deltas.

The important feature of this design is pairing before aggregation. Matching means after averaging could hide seed-level failures. Paired seed-level and epoch-level comparison tests whether the solver branch changes the stochastic infection trajectory itself.

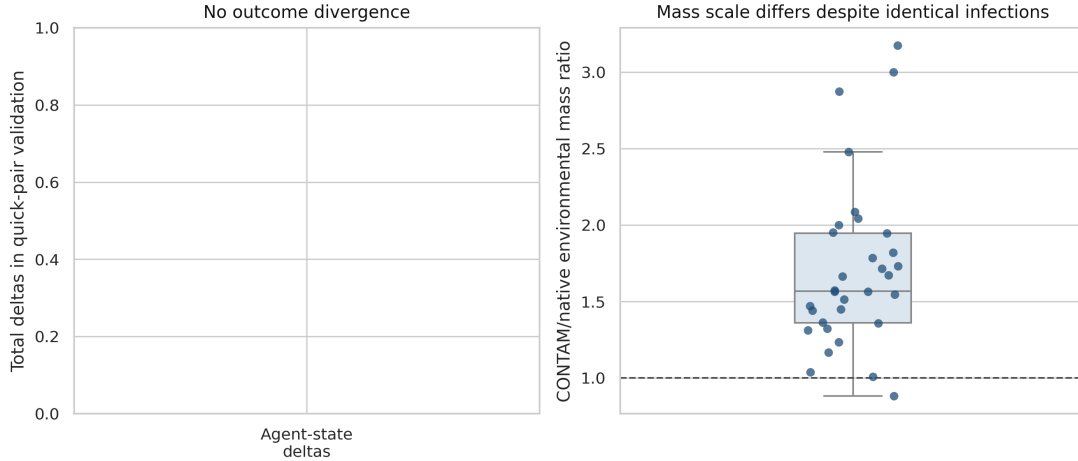


Figure 6.1: Native-versus-CONTAM quick-pair validation. Agent-state deltas were zero in the 40-run quick-pair telemetry check, while CONTAM/native environmental mass ratios differed, with median ratio 1.57 among finite ratios.

6.3 Environmental mass differs

The same validation shows that environmental pathogen mass is not identical. In the quick-pair file, the finite CONTAM/native median mass ratio has median 1.57, interquartile range 1.36–1.95, and maximum 3.17. Project notes summarize broader mass differences of roughly 1.3–1.7-fold. These differences did not change infection outcomes in the norovirus validation because the calibrated dose-response and route mixture made the infection process insensitive to that mass-scale difference in the tested regime.

This result is a solver-validation success for attack-rate experiments and a warning for environmental surveillance. If wastewater, air, or surface-swab triggers use absolute concentration thresholds, native and CONTAM mass scales must be calibrated separately. Neither simulated mass scale should be treated as empirical ground truth without external concentration data.

6.4 Scope of validation

The available validation is strongest for the norovirus calibration case and for infection outcomes. It supports using the native solver in large intervention sweeps where attack rate, outbreak occurrence, and timing are the primary outcomes. It does not prove equivalence for every respiratory pathogen, every HVAC configuration, or every environmental sampling protocol. Respiratory validation with SARS-CoV-2 and measles remains necessary before solver choice can be ignored for aerosol-dominant questions [11, 33, 41, 42].

Chapter 7

Detection and Intervention Results

7.1 Claim: onboard surveillance saturates after the first effective response trigger

The intervention experiments show a consistent pattern: once any effective onboard response is active, most of the available attack-rate reduction has already been captured. More elaborate mid-voyage surveillance can shift response timing, but it rarely changes final attack rate by much when syndromic detection and threshold escalation are already operating.

This is not an argument against wearables, wastewater, or multiplex diagnostics. It is an argument about where their highest value is likely to sit. In these runs, their mid-outbreak value is small because the ship is already responding. Their larger value is more likely at boundaries: pre-boarding risk detection, terminal wastewater, turnaround cleaning decisions, early cancellation, port-call modification, or fleet-level situational awareness.

7.2 Basic syndromic response captures most benefit

For the susceptible or baseline-immunity strata used in the main comparisons, no-response mean attack rates are high: 17.2% for norovirus, 20.7% for SARS-CoV-2, 37.9% for measles in a susceptible population, 66.1% for influenza, and 12.0% for *C. difficile*. Basic syndromic response reduces those means to 5.9%, 0.4%, 1.3%, 1.9%, and 0.2%, respectively. The absolute reductions are 11.3 percentage points for norovirus, 20.3 points for SARS-CoV-2, 36.6 points for measles, 64.2 points for influenza, and 11.8 points for *C. difficile*.

Figure 7.1 shows the full pattern. The first contrast, from no response to syndromic response, is much larger than contrasts among active surveillance strategies. For fast respiratory pathogens, even a small reduction in response time can matter early, but once the response is active, the residual outbreak size is already near the floor in the tested configuration.

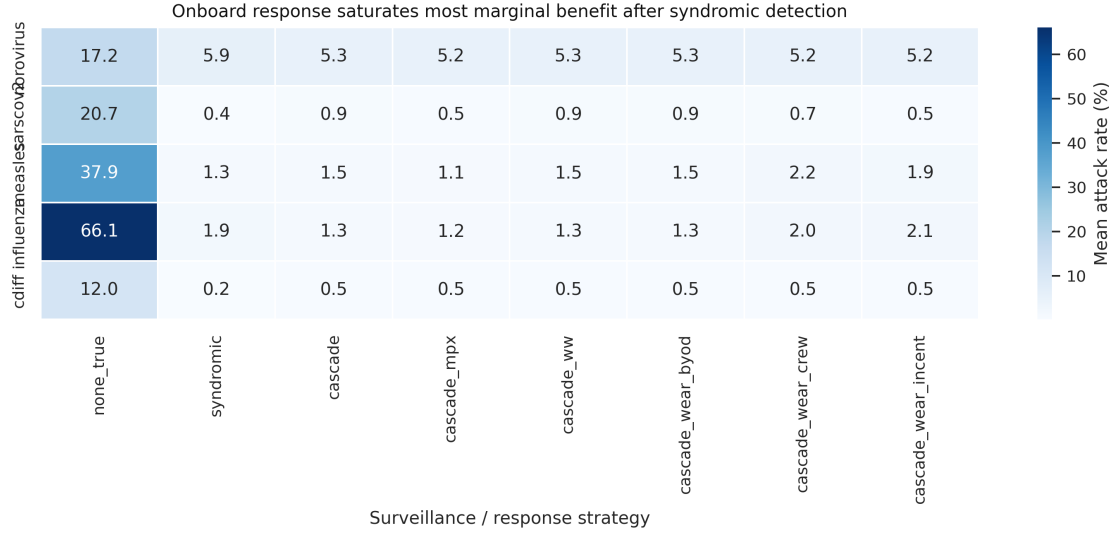


Figure 7.1: Mean attack rates by pathogen and response strategy for the baseline susceptibility strata. No-response attack rates collapse once syndromic or cascade response is active. Values are percent attack rate.

7.3 Wearables, wastewater, and multiplex testing add small mid-voyage gains

Norovirus provides the clearest example because it is the calibrated VSP pathogen. In the baseline-immunity stratum, no-response attack rate averages 17.2%. Syndromic response reduces it to 5.9%. Cascade strategies produce means between 5.2% and 5.3% for the strongest active-response settings in the same stratum. The marginal improvement over syndromic response is therefore less than one percentage point.

Respiratory pathogens show the same saturation in a lower residual range. SARS-CoV-2 falls from 20.7% with no response to 0.4% with syndromic response and 0.5–0.9% under cascade variants. Measles in a susceptible population falls from 37.9% to 1.3% under syndromic response and to 1.2–2.2% under cascade variants. Influenza falls from 66.1% to 1.9% under syndromic response and to 1.2–2.1% under cascade variants. These differences are small compared with the first response trigger and within the stochastic variability expected for shipboard simulations.

The literature on wearable and wastewater surveillance supports plausibility for earlier detection but does not guarantee final-outcome superiority after a response process is already active [43–52]. Rapid molecular diagnostics can improve clinical specificity and accelerate pathogen identification [53–55], but attack-rate benefit depends on whether that information changes control actions before the outbreak has already been contained by syndromic protocols.

7.4 The VSP threshold dominates medical knob sensitivity

The norovirus threshold sweep in Chapter 5 identifies the high-leverage policy parameter: the response threshold. In contrast, the norovirus medical-sensitivity sweep showed little final attack-rate movement across sick-call probability, detection delay, and isolation-compliance settings once the threshold backstop was active. This result is mechanistic. Moderate changes in individual reporting behavior matter less if the cumulative symptomatic fraction inevitably crosses a regulatory escalation threshold early in the outbreak.

The policy implication is narrow but useful. Improving compliance and sick-call behavior remains desirable for clinical care, passenger welfare, and legal duty of care. It may also matter in low-incidence or boundary-detection regimes. In the calibrated mid-voyage norovirus regime, however, final attack rate is more sensitive to when the ship changes operating mode than to moderate variation in the path by which the first few symptomatic cases enter the clinic.

7.5 Boundary surveillance is the likely return-on-investment target

The model points toward a different use case for advanced surveillance. If onboard response saturates quickly after symptomatic recognition, the remaining preventable burden lies before the onboard outbreak becomes established. Wearable pre-boarding data could identify symptomatic or pre-symptomatic passengers before boarding. Terminal wastewater or shore-side building wastewater could detect outbreak seeding before passengers board. Turnaround environmental sampling could identify persistent contamination. Fleet-level anomaly detection could alter staffing, cleaning, testing, and itinerary decisions before the voyage enters the mid-outbreak regime.

These boundary uses are not directly tested in the current campaigns because the itinerary and port layers are implemented as configuration stubs in the available runs. They should be the next intervention campaigns rather than another mid-voyage surveillance stack.

Chapter 8

Starfleet Generalization

8.1 Claim: the framework generalizes, but the demonstration is secondary

The Starfleet scenarios are a demonstration of arbitrary-platform generalization. They are not the core scientific evidence for the cruise-ship claims. Their value is that they stress the model outside the cruise setting while preserving the same conceptual elements: confined population, platform topology, medical response, route-weighted pathogens, and intervention timing.

The two platforms are a smaller Constitution-class ship with 430 agents and a larger Galaxy-class ship with 1,014 agents. Scenario pathogens include Rigelian fever, Psi 2000 polywater intoxication, Barclay protomorphosis, and a cross-era influenza comparison. Surveillance strategies include no response, sickbay response, Starfleet medical response, and, on Galaxy-class runs, combadge bioscan response.

8.2 Topology and response interact

Uncontrolled starship scenarios often saturate. Rigelian fever, Psi 2000, and Constitution-class influenza reach 100% mean attack rate in no-response runs. Medical response reduces them substantially: Constitution-class Rigelian fever falls to 25.7–27.2%, Psi 2000 to 27.7–27.8%, and influenza to 33.3–34.9%. On the larger Galaxy platform, Barclay protomorphosis falls from 100% under no response to roughly 49.7–51.8% under response, and influenza falls from 85.0% to roughly 40.0–41.6%.

The smaller Constitution platform can contain several response scenarios better than the larger Galaxy platform despite less advanced nominal surveillance. That pattern echoes the cruise findings: platform size, topology, and response timing can dominate technology labels. More advanced sensors do not automatically outperform simpler response if they do not change action early enough.

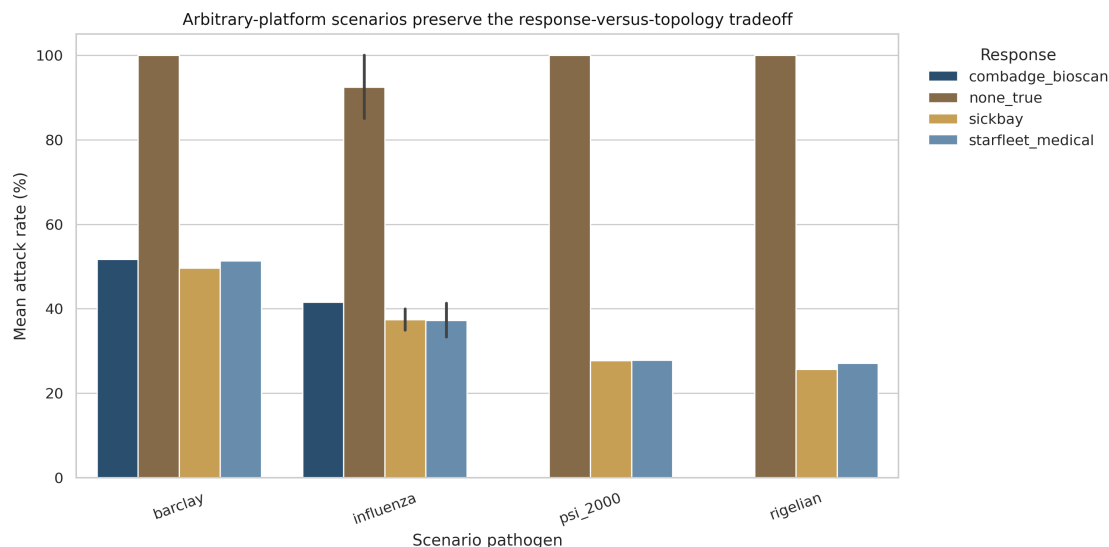


Figure 8.1: Starfleet scenario outcomes. Arbitrary-platform runs preserve the main model logic: confined topology can support saturation, while medical response substantially reduces final attack rate.

8.3 Interpretive limits

These scenarios use stylized fictional pathogens and platforms. They should not be interpreted as biological calibration. They demonstrate software generality and conceptual portability: if a confined environment can be represented as agents, zones, routes, and response triggers, CTB can evaluate the interaction of topology and intervention. The cruise-ship chapters supply the empirical and policy-relevant evidence.

Chapter 9

Discussion

9.1 Main interpretation

The CTB results support three linked conclusions. First, cruise-ship outbreak observations, especially VSP norovirus records, are intervention-shadowed measurements. They should not be treated as uncontrolled epidemic targets. Second, a single calibrated norovirus shedding parameter can preserve physical platform differences before response while matching the observed 5–7% VSP band after response. Third, onboard surveillance exhibits diminishing returns after the first effective response trigger. The highest-value surveillance questions are likely at voyage boundaries, not in the middle of an already recognized outbreak.

These conclusions are compatible with the maritime outbreak literature. VSP reporting, norovirus control, Diamond Princess, and other shipboard outbreaks all show that observation, behavior, policy, and pathogen biology are entangled [1–5, 9, 10, 15, 16]. CTB makes that entanglement explicit by modeling detection and response as part of the system.

9.2 Calibration philosophy

The VSP-shadow result changes how calibration should be judged. A model that matches observed VSP attack rates under `none_true` conditions is not necessarily well calibrated. It may be forcing an intervention-shadowed target into an uncontrolled biology parameter. The correct target depends on the observation process. For VSP data, that process includes a threshold and response. For a serosurvey, it would include sampling and test performance. For wastewater, it would include shedding, sewer mixing, assay limits, and concentration scaling. For shipboard clinic data, it would include sick-call behavior and medical-record ascertainment.

This principle matters beyond norovirus. Overdispersion, response timing, and observation bias are common in infectious-disease systems [25–27, 40]. Mechanistic models should therefore include the measurement process whenever the empirical target is produced by policy or behavior.

9.3 Implications for ship operations

The model suggests that VSP-like thresholded escalation is effective but blunt. It compresses attack rates across platforms, but it activates after symptomatic burden has accumulated. Lower thresholds reduce attack rate further, but they may increase operational costs, passenger disruption, and false escalation. The threshold is therefore a policy parameter with measurable epidemiological leverage and operational tradeoffs.

Dining operations remain a plausible structural lever. Self-service buffets, shared utensils, open seating, crew-mediated service, and food-contact sanitation can change exposure opportunities before medical response begins. The current campaigns parameterize dining at the class level, but future FOH and BOH crew segmentation is needed to test galley-mediated amplification, crew reservoirs, and targeted food-service screening.

9.4 Implications for surveillance technology

The intervention results do not support large claims for mid-voyage wearable or wastewater benefit once syndromic response is active. This conclusion is conditional on the simulated decision rules. If a cruise line used early wearable anomalies or wastewater signals to alter operations before syndromic cases crossed a response threshold, the result could change. That is exactly why boundary campaigns are the next step.

The CONTAM validation adds a second surveillance caveat. Infection outcomes can match across transport solvers while environmental mass differs by roughly 1.3–1.7-fold. Environmental surveillance thresholds must therefore be calibrated against empirical concentrations or solver-specific telemetry, not inferred from attack-rate validation alone.

9.5 Limitations

The available artifact set does not include full code-level pathogen configuration files, complete platform zone inventories, or raw per-agent telemetry for every campaign. The supplement therefore reports the real tables supported by supplied CSVs and context documents, and it flags missing exports rather than reconstructing unsupported constants. Port visits and boarding effects are specified but stubbed in the main runs. Respiratory CONTAM validation remains incomplete relative to the norovirus validation. Compliance effects may differ in parameter regimes where the VSP backstop does not activate or activates late.

The VSP dataset is also an imperfect empirical target. Passenger-count bins are uneven; the under-500 and above-5000 bins are sparse. Public VSP records select reported outbreaks and do not observe subthreshold events. The model result should be interpreted as a coherent explanation of the available data and campaign outputs, not as a universal law for every voyage.

Chapter 10

Future Work

10.1 Port visits and shore-excursion mixing

The itinerary layer should be activated in experiments that vary port-day passenger removal, shore-excursion mixing, re-boarding timing, and shore-acquired infection risk. Port days can reduce onboard density while introducing external exposure. Their net effect is not obvious without simulation.

10.2 Boarding and turnaround screening

The strongest surveillance hypothesis is now a boundary hypothesis. Future campaigns should test health declarations, symptom screening, rapid testing, terminal wastewater, wearable pre-boarding data, and boarding holds. Turnaround experiments should carry environmental reservoirs, crew infection states, and cleaning effectiveness between voyages.

10.3 Inter-voyage persistence

Norovirus and *C. difficile* motivate multi-voyage persistence models. A realistic model should carry galley contamination state, cabin and public-surface reservoirs, asymptomatic crew infections, and turnaround sanitation across sailings. VSP observations may reflect not only within-voyage transmission but also learning and control across repeat voyages.

10.4 Environmental surveillance calibration

Air, wastewater, and surface-swab thresholds need empirical calibration. The native and CONTAM solvers can agree on attack-rate outcomes while disagreeing on mass scale. Concentration data from

wastewater solids, terminal sampling, cabin air, or surface swabs should be used to set thresholds for each solver and sampling design [49–52].

10.5 FOH and BOH crew segmentation

Food-service transmission requires explicit front-of-house and back-of-house crew roles. Future models should separate galley preparation, buffet service, main-dining-room waitstaff, cabin stewards, medical staff, and general hotel operations. That segmentation would allow targeted crew screening, sick-leave policy tests, and galley-mediated outbreak scenarios.

10.6 Direct empirical validation

The model should be tested against datasets beyond public VSP summaries: line-list outbreak reports, daily clinic counts, sanitation logs, dining attendance telemetry, wearable occupancy data, wastewater concentrations, and environmental swabs. Such datasets would allow calibration of time-to-detection, not only final attack rate.

Appendix A

Supplemental Tables and Source Provenance

This supplement contains the tables used to support the monograph narrative. Internal campaign identifiers are retained here as provenance labels rather than as the organizing structure of the main text. The source data bundle was uploaded as `data_entry:ctb-monograph-v2-source-provenance-bundle-zip-hs39`.

Table A.1: Platform characteristics and operational assumptions used for interpretation.

Platform	Agents	Structural interpretation	Dining exposure	Medical response default	Boundary status
expedition	450	Small vessel; high staff-to-guest ratio; minimal unmonitored self-service	Buffet/casual 1.60; MDR 1.35; specialty 0.05 meals per passenger-day	sick-call 0.95; compliance 0.95; delay 0 epochs	Port and boarding hooks specified; inactive in main runs
classic	1,910	Premium/classic cruise vessel; mixed fixed and open seating	Buffet/casual 1.55; MDR 1.05; specialty 0.40	sick-call 0.70; compliance 0.90; delay 2 epochs	All-sea-day main campaign interpretation
spirit	3,000	Contemporary cruise vessel; high lido throughput and hybrid dining	Buffet/casual 1.70; MDR 1.08; specialty 0.22	sick-call 0.50; compliance 0.85; delay 4 epochs	All-sea-day main campaign interpretation
mega	7,000	Large distributed-venue vessel; high passenger anonymity; many food outlets	Buffet/food hall 1.40; MDR 0.80; specialty/casual 0.80	sick-call 0.30; compliance 0.80; delay 8 epochs	All-sea-day main campaign interpretation

Table A.2: Working pathogen-profile summary supported by campaign outputs and context documents.

Pathogen	Class	Dominant route emphasis	Dose values in supplied campaigns	Immunity strata	Use in monograph
norovirus	GI virus	fomite, food-service, direct contact, environmental	10.6 main calibration	0.0 and 0.2	Central VSP-shadow calibration
SARS-CoV-2	respiratory virus	short-range respiratory, aerosol/HVAC, contact	6.5 in main surveillance runs	0.0 and 0.3	Respiratory intervention comparison
influenza	respiratory virus	short-range respiratory, contact, aerosol	5.5	0.0	Fast respiratory response-stress case
measles	respiratory virus	aerosol/HVAC and respiratory contact	5.0	0.0 and 0.85	High-transmissibility airborne comparison
Clostridioides difficile	spore-forming bacterial GI pathogen	environmental reservoir, fomite, contact	5.0	0.0	Persistent environmental-pathogen comparison
campylobacter	foodborne bacterium	food-service and direct GI exposure	varied in broad sweeps	0.0	Route-weight breadth test
vibrio	food/water-associated bacterium	food and water exposure	varied in broad sweeps	0.0	Route-weight breadth test
legionella	environmental bacterium	aerosolized water and environmental exposure	varied in broad sweeps	0.0	Environmental route test
Ebola	high-consequence viral disease	close contact and bodily-fluid route proxy	varied in broad sweeps	0.0	Rare-pathogen stress test
hantavirus	high-consequence viral disease	respiratory/contact assumptions for Andes-like scenario	varied in broad sweeps	0.0	Rare shipboard scenario context

Table A.3: Surveillance and response strategy definitions.

Strategy	Definition	Interpretive use
none_true	No response; transmission proceeds without modeled detection or intervention	Baseline latent epidemic potential
syndromic	Symptomatic reporting plus stock VSP-like response at 3% GI threshold where applicable	Primary practical onboard response
synd_vsp_off	Sick-call response without VSP threshold back-stop	Separates medical reporting from regulatory threshold

Strategy	Definition	Interpretive use
synd_vsp5	VSP-like threshold at 5% passenger GI attack rate	Higher response threshold
synd_vsp1	VSP-like threshold at 1% passenger GI attack rate	Lower response threshold
cascade	Escalating response logic beyond basic syndromic detection	Base advanced-surveillance stack
cascade_mpx	Cascade with multiplex point-of-care diagnostic signal	Diagnostic-specific strategy
cascade_ww	Cascade with wastewater signal	Environmental surveillance strategy
cascade_wear_byod	Cascade with passenger bring-your-own-device wearable coverage	Passenger wearable strategy
cascade_wear_crew	Cascade with crew wearable coverage	Crew wearable strategy
cascade_wear_incent	Cascade with incentivized wearable coverage	Higher-adoption wearable strategy

Table A.4: Campaign manifest summary. Campaign IDs are source provenance and are not used as the main narrative structure.

Campaign	Runs	Purpose	Pathogens	Platforms
C7	5250	Three-platform norovirus and SARS-CoV-2 recalibration	norovirus; SARS-CoV-2	expedition; spirit; classic;
C8	2295	Ten-pathogen route-weight sweep	10 real-world pathogens	expedition; spirit; classic;
C9	8745	Fine calibration and density-exponent sweep	7 dose-responsive pathogens	expedition; spirit; classic;
C10	1140	Expedition dining-fix verification	norovirus; SARS-CoV-2	expedition
C12	6720	Four-class recalibration with per-class meal weights	7 dose-responsive pathogens	expedition; spirit; mega; classic;
C12b	3900	Syndromic calibration sweep	norovirus; SARS-CoV-2	4 cruise classes
C12c	5920	Fine norovirus calibration and medical-response sensitivity	norovirus	4 cruise classes
C14	8440	Advanced surveillance and VSP-threshold sweep	norovirus; SARS-CoV-2; influenza; measles; C. difficile	4 cruise classes
C14b	4220	Doubled small-vessel surveillance coverage	5 pathogens	expedition; classic

Campaign	Runs	Purpose	Pathogens	Platforms
Trek	5520	Arbitrary-platform demonstration	demon- 4 scenario pathogens	Constitution; Galaxy

Table A.5: Parsed VSP norovirus passenger attack rates by passenger-count bin.

Passenger bin	Mean AR (%)	Median AR (%)	Outbreaks	IQR (%)
< 500	17.94	11.98	4	7.62–22.30
500-1500	6.15	5.09	70	3.63–6.87
1500-3000	5.91	5.22	111	4.25–6.68
3000-5000	6.50	5.78	30	4.90–7.80
> 5000	8.91	8.91	1	8.91–8.91

Table A.6: Norovirus calibration at dose_adj=10.6. Response-class means aggregate the available C12c seeds and response variants.

Platform	Response class	Mean AR (%)	95% CI	Runs
expedition	none_true	23.42	17.16–29.68	50
expedition	syndromic/VSP variants	6.65	6.07–7.24	490
expedition	stock syndromic	6.26	4.48–8.05	50
classic	none_true	22.62	16.84–28.39	50
classic	syndromic/VSP variants	7.10	6.52–7.68	490
classic	stock syndromic	7.04	5.19–8.89	50
spirit	none_true	12.18	9.22–15.14	50
spirit	syndromic/VSP variants	4.60	4.17–5.04	490
spirit	stock syndromic	5.08	3.64–6.52	50
mega	none_true	10.63	7.50–13.76	50
mega	syndromic/VSP variants	5.52	4.88–6.16	490
mega	stock syndromic	5.09	3.20–6.98	50

Table A.7: Selected intervention outcomes in the baseline-susceptibility stratum of the C14 surveillance sweep.

Pathogen	Strategy	Mean AR (%)	95% CI	Runs
norovirus	none_true	17.21	14.71–19.72	200
norovirus	syndromic	5.87	4.99–6.74	200
norovirus	cascade_mpx	5.23	4.13–6.33	120
norovirus	cascade_ww	5.30	4.17–6.42	120
norovirus	cascade_wear_crew	5.23	4.13–6.34	120
norovirus	cascade_wear_incent	5.24	4.14–6.33	120
sarscov2	none_true	20.72	15.87–25.57	120

Pathogen	Strategy	Mean AR (%)	95% CI	Runs
sarscov2	syndromic	0.38	0.04–0.73	120
sarscov2	cascade_mpx	0.49	-0.03–1.00	120
sarscov2	cascade_ww	0.86	0.01–1.71	120
sarscov2	cascade_wear_crew	0.74	0.05–1.43	120
sarscov2	cascade_wear_incent	0.51	-0.04–1.07	120
measles	none_true	37.91	31.52–44.30	120
measles	syndromic	1.32	-0.14–2.77	120
measles	cascade_mpx	1.15	-0.12–2.42	120
measles	cascade_ww	1.47	0.05–2.88	120
measles	cascade_wear_crew	2.17	0.21–4.13	120
measles	cascade_wear_incent	1.88	0.06–3.70	120
influenza	none_true	66.08	58.27–73.89	120
influenza	syndromic	1.89	-0.40–4.18	120
influenza	cascade_mpx	1.20	-0.43–2.83	120
influenza	cascade_ww	1.28	-0.37–2.93	120
influenza	cascade_wear_crew	2.04	-0.15–4.23	120
influenza	cascade_wear_incent	2.14	-0.17–4.45	120
cdiff	none_true	12.03	8.09–15.96	120
cdiff	syndromic	0.18	0.14–0.23	120
cdiff	cascade_mpx	0.54	-0.19–1.28	120
cdiff	cascade_ww	0.53	-0.18–1.25	120
cdiff	cascade_wear_crew	0.55	-0.19–1.29	120
cdiff	cascade_wear_incent	0.54	-0.18–1.26	120

Table A.8: Native-versus-CONTAM validation summary.

Validation component		Scale	Result
Broad paired validation		200 paired runs	0 deltas in attack rate, peak prevalence, outbreak occurrence, timing, R-effective, quarantine burden, final susceptible fraction
Per-seed	per-epoch	6,720 agent-state observations	6,720 exact matches; 0 mismatches
norovirus validation	telemetry	40 paired runs	0 total agent-state deltas
Quick-pair	telemetry	40 paired runs	0 total agent-state deltas
recheck			
Environmental mass ratio		32 finite quick-pair ratios	median 1.57; IQR 1.36–1.95; maximum 3.17

Table A.9: Starfleet scenario summary. These rows are a demonstration of arbitrary-platform generalization, not cruise calibration evidence.

Pathogen	Platform	Response	Mean AR (%)	Runs
barclay	galaxy	combadge_bioscan	51.75	240
barclay	galaxy	none_true	100.00	240
barclay	galaxy	sickbay	49.68	240
barclay	galaxy	starfleet_medical	51.34	240
influenza	constitution	none_true	100.00	180
influenza	constitution	sickbay	34.91	180
influenza	constitution	starfleet_medical	33.27	180
influenza	galaxy	combadge_bioscan	41.57	600
influenza	galaxy	none_true	85.00	600
influenza	galaxy	sickbay	39.95	600
influenza	galaxy	starfleet_medical	41.26	600
psi_2000	constitution	none_true	100.00	240
psi_2000	constitution	sickbay	27.72	240
psi_2000	constitution	starfleet_medical	27.82	240
rigelian	constitution	none_true	100.00	300
rigelian	constitution	sickbay	25.69	300
rigelian	constitution	starfleet_medical	27.15	300

Table A.10: Raw dataset and context-document provenance. The user-facing source bundle is referenced in the supplement introduction.

Source	Rows	Columns	Use
c7_campaign_results.csv	5250	14	campaign aggregate outcomes
c8_campaign_results.csv	2295	12	campaign aggregate outcomes
c9_campaign_results.csv	8745	11	campaign aggregate outcomes
c10_campaign_results.csv	1140	9	campaign aggregate outcomes
c12_campaign_results.csv	6720	13	campaign aggregate outcomes
c12b_campaign_results.csv	3900	10	campaign aggregate outcomes
c12c_campaign_results.csv	5920	9	campaign aggregate outcomes
c14_campaign_results.csv	8440	10	campaign aggregate outcomes
c14b_campaign_results.csv	4220	10	campaign aggregate outcomes
trek_campaign_results.csv	5520	10	campaign aggregate outcomes
vsp_outbreak_data.csv	285	5	parsed VSP outbreak records
Dining model data report	document	–	dining and food-service parameterization
Medical care and staffing report	document	–	medical response and staffing parameterization
ship_operations_spec.md	document	–	voyage/port-layer specification
medical_response_spec.md	document	–	medical-response parameters

Appendix B

Known Missing Exports

The supplied artifact set does not contain full code-level route-weight configuration files, complete platform zone inventories, raw per-agent telemetry for every campaign, or empirical environmental concentration data. The main text and supplement therefore report only values supported by the campaign CSVs, context documents, or validation summaries. A future synchronized build should export pathogen configuration JSON, zone inventories by platform and zone type, FOH/BOH crew role counts, and solver-specific environmental concentration calibration tables.

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